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ROLL NUMBER:

Autumn 2025
IIT Bombay

MA 403 Real Analysis

Quiz 1

1. **(2 marks)** Let S be a nonempty subset of $\mathbb{R}$, which is bounded above. Define $\sup S$.
2. **(2 marks)** Let S be a nonempty subset of $\mathbb{R}$ such that $\sup S$ exists. Show that $\sup S$ is uniquely determined.

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Quiz 2

1. **(2 marks)** Let $f : S \rightarrow T$ be a one-to-one function. Prove that $f^{-1}(f(A)) = A$ for every subset A of S .
2. **(2 marks)** Let S be the set of all circles in $\mathbb{R}^2$ having integer radii and centers with integer coordinates. Is S a countable set? Explain your answer.

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Quiz 3

1. **(1 mark)** Define accumulation point of a set S .
2. **(1 mark)** Define closure of a set S .
3. Consider $S = \{(x, y) : x \geq 0\} \subset \mathbb{R}^2$.
(1 mark) Determine all accumulation points of S in $\mathbb{R}^2$.
(1 mark) Explain whether S is an open set or a closed set (or neither).

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Quiz 4

1. **(1 mark)** Give an example of a non-compact subset S of $\mathbb{R}$, and an example of an open cover for this S , which has no finite subcover.
(Just state the examples, don't give a proof.)
3. **(2 marks)** For $\mathbf{x} = (x_1, \dots, x_n)$ and $\mathbf{y} = (y_1, \dots, y_n)$ in $\mathbb{R}^n$, define $d(\mathbf{x}, \mathbf{y}) = \max_{1 \leq i \leq n} |x_i - y_i|$.
Prove that d is a metric on $\mathbb{R}^n$.
4. **(1 mark)** Consider the metric space (M, d) where $M = \mathbb{Z}$ and $d(x, y) = |x - y|$. Prove that in this metric space, every subset S is open.

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Quiz 5

1. **(1 mark)** Let (M, d) be a metric space, and let S, T be subsets of M such that $S \subset T$.
Prove that $\overline{S} \subset \overline{T}$.

2. **(3 marks)** Let $f : [0, 1] \rightarrow \mathbb{R}$ be given by $f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational} \\ x & \text{if } x \text{ is rational.} \end{cases}$

Prove that f is continuous at $x = 0$, and is not continuous at other points.

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Quiz 6

1. (2 marks) Let $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$

Prove that f is continuous at 0.

Prove that f is not differentiable at 0.

2. (2 marks) Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and assume that f is differentiable at all points in (a, b) , and suppose $f'(x) = 0$ for all $x \in (a, b)$. Prove that f is a constant function.

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Quiz 7

- (2 marks)** For the function $f(x) = \sin x$, find a polynomial P such that P approximates f to order 2 at the point $a = 0$. Justify your answer.
- (2 marks)** Let f be an increasing function defined on $[a, b]$ and let $x_1, \dots, x_n$ be n points in (a, b) such that $a < x_1 < x_2 < \dots < x_n < b$.

Show that
$$\sum_{k=1}^n [f(x_k+) - f(x_k-)] \leq f(b) - f(a).$$

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Quiz 8

1. (2 marks) Let $f : [a, b] \rightarrow \mathbb{R}$ be a function satisfying

$$|f(x) - f(y)| < 10|x - y| \quad \text{for all } x \text{ and } y \text{ in } [a, b].$$

Prove that f is of bounded variation on $[a, b]$.

2. Let $f : [0, 1] \rightarrow \mathbb{R}$ be the function defined by: $f(x) = \begin{cases} 1 & \text{if } x = \frac{1}{2} \\ 0 & \text{otherwise.} \end{cases}$

(1 mark) Prove that f is Riemann-integrable.

(1 mark) Prove that $\int_0^1 f(x)dx = 0$.

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Quiz 9

Suppose $g : [a, b] \rightarrow \mathbb{R}$ is a bounded function which is Riemann-integrable on $[a, b]$, and $f : [a, b] \rightarrow \mathbb{R}$ is a function such that $f(x) = g(x)$ except at a finite number of points x .

1. **(2 marks)** Prove that f is Riemann-integrable.
2. **(2 marks)** Prove that

$$\int_a^b f(x)dx = \int_a^b g(x)dx.$$

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Quiz 10

1. **(2 marks)** Let $f : [0, 1] \rightarrow \mathbb{R}$ be bounded and Riemann-integrable on $[0, 1]$. Show that the improper integral of f on $[0, 1]$ exists and equals the Riemann integral of f on $[0, 1]$.
2. **(1 mark)** Give an example of real sequences $\{a_n\}$ and $\{b_n\}$ for which

$$\limsup_{n \rightarrow \infty} (a_n + b_n) < \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n.$$

3. **(1 mark)** Explain what is meant by convergence of a series $\sum_{n=1}^{\infty} a_n$, where $a_n \in \mathbb{R}$ for each $n \in \mathbb{Z}_+$.

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Quiz 11

1. **(2 marks)** Explain what it means for a sequence of functions $\{f_n\}$ to converge uniformly.
2. **(2 marks)** Prove that a uniformly convergent sequence of bounded functions $\{f_n\}$ is uniformly bounded.